3.1 Introduction to Python

Definition

Python ek **high-level, interpreted, general-purpose programming language** hai jo readable syntax aur easy programming ke liye famous hai.

Python ko **Guido van Rossum** ne **1989** me develop kiya aur **1991** me release kiya.

Python ka naam TV comedy show **"Monty Python's Flying Circus"** se inspire hai.

Python ko beginner language kyu bolte hain?

Python me English jaisi syntax hoti hai.

Isliye code likhna aur samajhna bahut easy hota hai.

Example

C Language

printf("Hello");

Python

print("Hello")

Python me semicolon aur braces ki zarurat nahi hoti.

Isi wajah se beginners Python ko jaldi sikh lete hain.

Real Life Example

Socho tum English me kisi dost ko message bhejte ho.

Python bhi almost waise hi readable language hai.

Jitna simple likhoge utna hi aasani se computer samajh lega.

3.1 Features of Python

Python ki sabse important features niche diye gaye hain.

---

1. Simple and Easy to Learn

Theory

Python ka syntax bahut simple hai.

Kam keywords hote hain aur code readable hota hai.

Isliye beginners ke liye best language mani jati hai.

Example

```
print("Welcome")
```

---

2. Interpreted Language

Theory

Python line by line execute hota hai.

Compiler ki tarah pehle pura program compile nahi hota.

Error jis line me hoti hai wahi dikh jati hai.

Advantage

✓ Debugging easy hoti hai.

---

3. Interactive Language

Python me Interactive Mode hota hai.

Isme turant code likho aur output dekh lo.

Example

>>>5+2

7

---

4. Object-Oriented

Python OOP concepts support karta hai.

Jaise

- Class

- Object
- Inheritance
- Polymorphism
- Encapsulation

Large software banane me help karta hai.

---

5. Portable

Ek baar code likho.

Windows

Linux

Mac

Sab jagah run kar sakta hai.

Isko Platform Independent bhi bolte hain.

---

6. Scalable

Python chhote program se lekar bade software tak bana sakta hai.

Examples

- Instagram
- YouTube

- Netflix
- Dropbox

---

7. Extensible

Python ko C aur C++ ke saath use kar sakte hain.

Performance badhane ke liye dusri languages ka code bhi add kar sakte hain.

---

8. Dynamic Typing

Python me datatype declare nahi karna padta.

Interpreter khud decide karta hai.

Example

a=10

b=3.5

c="Vijay"

Yaha

a → int

b → float

c → string

Automatically ban gaye.

---

9. GUI Programming Support

Python se Graphical Applications bana sakte hain.

Popular Libraries

- Tkinter
- PyQt
- Kivy

---

10. Large Standard Library

Python ke paas bahut saare built-in modules hain.

Jaise

math

random

os

time

datetime

In modules ki wajah se coding easy ho jati hai.

Python Features Shortcut

Simple

Interpreted

Interactive

Object-Oriented

Portable

Scalable

Extensible

Dynamic

GUI Support

Large Library

Exam me ye sequence yaad rakhna.

3.2 How to Run Python

Python chalane ke teen tarike hain.

Method 1

Interactive Interpreter

Is mode me line by line code execute hota hai.

Steps

Open Python

>>>

Likho

print("Hello")

Output

Hello

Kab use karte hain?

Practice

Testing

Learning

Method 2

Script Mode

Program ko file me save karte hain.

Extension

.py

Example

hello.py

Code

```
print("Hello Vijay")
```

Run

```
python hello.py
```

Output

```
Hello Vijay
```

Kab use karte hain?

Real Projects

Assignments

Applications

Method 3

IDLE (Integrated Development and Learning Environment)

Python ke saath aane wala editor.

Features

✓ Code likho

✓ Run karo

✓ Save karo

✓ Edit karo

Beginners ke liye best hai.

Difference

Interactive	Script
Line by Line	Pura Program
Practice	Projects
Temporary	Permanent

3.3 Identifiers

Definition

Identifier kisi bhi

- Variable
- Function
- Class
- Object

ka naam hota hai.

Example

age=18

Yaha

age

Identifier hai.

Rules of Identifier

Rule 1

Letter ya underscore (_) se start hona chahiye.

✓ Correct

age

_name

Student

❌ Wrong

2age

Rule 2

Number se start nahi kar sakte.

❌

12abc

Rule 3

Special Symbols allowed nahi.

❌

sum@

age#

salary%

Rule 4

Keywords use nahi kar sakte.

❌

if=10

Rule 5

Case Sensitive

age

Age

AGE

Teen alag identifiers hain.

Naming Convention

Variable

student_name

Function

calculate()

Class

Student

Private Variable

_marks

Strong Private

__salary

Special Python Variable

__name__

Valid Identifiers

marks

student

student_name

_roll

Student

Invalid Identifiers

2marks

class

roll#

sum%

name@

Real Life Example

Socho school me har student ka ek naam hota hai.

Waise hi programming me har variable ka ek naam hota hai.

Isi naam ko Identifier bolte hain.

3.4 Reserved Keywords

Definition

Keywords wo special words hote hain jinka meaning Python ne pehle se fix kar diya hai.

Inhe variable ke naam ki tarah use nahi kar sakte.

Example

if

else

while

for

class

return

import

Wrong Example

if=20

Error aayega.

Correct Example

marks=20

Keywords yaad rakhne ki Trick

Keywords Python ke commands hain.

Identifiers tumhare diye hue names hote hain.

3.5 Variables

Definition

Variable ek memory location hota hai jisme data store hota hai.

Simple words me

Variable = Data Store Karne Ka Box

Example

name="Vijay"

age=18

marks=90

Diagram

Variable

↓

Memory

↓

Value

Variable Assignment

a=10

Yaha

a

Variable hai.

10

Value hai.

Multiple Assignment

a=b=c=100

Teen variables me same value.

Multiple Variables

a,b,c=10,20,30

Output

print(a)

Output

10

Variable Naming Tips

✓ Meaningful Name

student_name

❌

x1

3.6 Comments in Python

Definition

Comment wo text hota hai jise programmer samjhane ke liye likhta hai.

Python Interpreter comments ko ignore karta hai.

Single Line Comment

This is comment

print("Hello")

Multi-Line Comment

'''

This

is

Comment

'''

Ya

"""

Python

Comment

"""

Why Comments?

✓ Code samajhne ke liye

✓ Team Work

✓ Future Revision

✓ Debugging

Real Life Example

Book me pencil se likhe notes ki tarah.

Book padhte waqt teacher us note ko explain karta hai.

Waise hi comments programmer ke notes hote hain.

3.7 Indentation in Python

Definition

Python me code block dikhane ke liye spaces ya tab ka use karna **Indentation** kehlata hai.

Dusri languages { } use karti hain, lekin Python **Indentation** use karta hai.

Correct Example

```
if True:
    print("Correct")
else:
    print("Wrong")
```

Wrong Example

```
if True:
```

print("Correct")

Output

IndentationError

Rule

- Generally **4 spaces** use ki jati hain.
- Ek block ke andar sabhi lines ki indentation same honi chahiye.
- Indentation galat hogi to program error dega.

Real Life Example

School me students line me khade hote hain. Agar koi line tod de to arrangement bigad jata hai. Usi tarah Python me indentation code ki arrangement hoti hai. Agar indentation galat hui to Python code ko samajh nahi pata.

📝 **Part 1 Quick Revision**

- Python ek high-level, interpreted language hai.
- Python ke 10 major features yaad rakho.

- Python run karne ke 3 methods: Interactive, Script aur IDLE.
- Identifier kisi variable, function ya class ka naam hota hai.
- Keywords reserved words hote hain, unhe identifier nahi bana sakte.
- Variable data store karta hai.
- Comments code ko samjhane ke liye hote hain.
- Python me block define karne ke liye indentation bahut important hai.

3.8 Multi-Line Statements

Definition

Python me normally **ek statement ek line me likha jata hai.**

Lekin agar statement bahut lamba ho to use **multiple lines** me likh sakte hain.

Isko **Multi-Line Statement** kehte hain.

Syntax 1 (Backslash \)

Agar statement ko next line me continue karna ho to **\ (backslash)** ka use karte hain.

Example

```
total = 10 + \
        20 + \
        30
print(total)
```

Output

60

Explanation

Python samajhta hai ki statement abhi complete nahi hua hai.

Isliye next line ko bhi same statement ka part maan leta hai.

Syntax 2 (Brackets)

Agar statement

- ()

- []
- {}

ke andar ho to backslash ki zarurat nahi hoti.

Example

```
months = [
   "January",
   "February",
   "March",
   "April"
]


print(months)
```

Output

```
['January','February','March','April']
```

Syntax 3 (Single Line Multiple Statements)

Ek hi line me multiple statements likh sakte hain.

Semicolon (;) ka use hota hai.

Example

a=10; b=20; c=30

print(a,b,c)

Output

10 20 30

Real Life Example

Socho tum ek bahut lambi application likh rahe ho.

Ek line me sab likhne se padhna mushkil hoga.

Isliye usko kai lines me likhte ho.

Python bhi isi tarah long statement ko multiple lines me likhne deta hai.

Interview Point

✓ Long statement ko readable banane ke liye Multi-Line Statements use kiye jate hain.

3.9 Multiple Statement Group (Suite)

Definition

Python me **same indentation wale statements ka group** ko **Suite** kehte hain.

Ye mostly use hota hai

- if
- else
- while
- for
- function
- class

ke andar.

Syntax

```
if condition:
    statement1
    statement2
    statement3
```

Yaha

Teen statements milkar

Ek **Suite** banate hain.

Example

marks=80

if marks>=33:

    print("Pass")

    print("Congratulations")

Output

Pass

Congratulations

Diagram

if condition:

↓

Statement 1

Statement 2

Statement 3

↓

Suite

Important Rule

Suite ki saari lines ki indentation same honi chahiye.

Wrong Example

```
if True:
    print("Hello")
  print("World")
```

Output

IndentationError

Real Life Example

School assembly me ek class ke saare students ek line me khade hote hain.

Ye group hi Suite jaisa hai.

Interview Point

Suite = Same indentation wale statements ka group.

3.10 Quotes in Python

Definition

Python me String likhne ke liye Quotes ka use hota hai.

Python teen tarah ke Quotes support karta hai.

---

1. Single Quotes

```
name='Vijay'
```

---

2. Double Quotes

```
city="Patna"
```

---

3. Triple Quotes

```
paragraph="""Python
is
easy"""
```

Output

Python

is

easy

Difference

Quotes	Use
' '	Small String
" "	Small String
''' '''	Multi-line String
""" """	Multi-line String

Example

```
a='Hello'
b="World"
c="""Python
Programming"""
print(a)
print(b)
```

print(c)

Real Life Example

Quotes waise hi hain jaise notebook me kisi sentence ko " " ke andar likhna.

Python samajh jata hai ki ye String hai.

3.11 Input, Output and Import Functions

Ye chapter ka sabse important topic hai.

Interview me bhi bahut pucha jata hai.

3.11.1 Output Function

print()

Definition

Screen par output dikhane ke liye

print()

function ka use hota hai.

Syntax

print(value)

Example 1

print("Hello")

Output

Hello

Example 2

a=10

print(a)

Output

10

Example 3

name="Vijay"

age=18

print(name,age)

Output

Vijay 18

---

print() Parameters

Book me diya syntax

print(objects,

sep=' ',

end='\n')

Isme do important cheezein hain.

---

sep Parameter

Separator decide karta hai.

Default separator

Space hota hai.

Example

print(1,2,3)

Output

1 2 3

Example

print(1,2,3,sep="+")

Output

1+2+3

Example

print("A","B","C",sep="-")

Output

A-B-C

---

end Parameter

Normally print ke baad cursor next line me chala jata hai.

Ye

end="\n"

ki wajah se hota hai.

Example

```
print("Hello",end=" ")


print("Vijay")
```

Output

Hello Vijay

Example

```
print("Python",end="*")


print("Programming")
```

Output

Python*Programming

Summary

Parameter	Work
sep	Separator change karta hai
end	Ending change karta hai

3.11.2 Input Function

Definition

User se data lene ke liye

input()

function use hota hai.

Syntax

input("Message")

Example

```
name=input("Enter Name : ")


print(name)
```

Output

```
Enter Name : Vijay


Vijay
```

Example

```
age=input("Enter Age : ")


print(age)
```

Output

```
Enter Age :18


18
```

Important Point

Python 3 me

input()

String return karta hai.

Agar number chahiye to

int()

use karna padta hai.

Example

```
age=int(input("Enter Age : "))
```

---

raw_input()

Book me

raw_input()

likha hai.

Ye sirf **Python 2** me hota tha.

Python 3 me ye **remove** kar diya gaya hai.

Ab sirf

input()

use hota hai.

Difference

Python 2	Python 3
raw_input()	input()

3.11.3 Import Function

Definition

Python ke kisi module ko program me use karne ke liye

import

keyword ka use hota hai.

Module Kya Hota Hai?

Module ek Python file hoti hai jisme pehle se bane functions hote hain.

Syntax

import module_name

Example

import math

print(math.pi)

Output

3.1415926535

Another Example

import random

print(random.randint(1,10))

Output

7

(Har baar alag number aa sakta hai.)

Common Modules

Module	Work
math	Mathematics
random	Random Number
time	Time
os	Operating System
datetime	Date & Time

Real Life Example

Socho tumhare paas toolbox hai.

Jab screwdriver chahiye tabhi usko toolbox se nikalte ho.

Python me bhi

import

toolbox se tools lane jaisa hai.

📝 Part 2 Quick Revision

- **Multi-Line Statement** → Long statement ko multiple lines me likhna.
- **Suite** → Same indentation wale statements ka group.
- **Quotes** → ' ', " ", ''' ''', """ """ strings ke liye use hote hain.
- **print()** → Output dikhata hai.
- **sep** → Values ke beech separator badalta hai.
- **end** → Print ke baad kya aayega, ye decide karta hai.
- **input()** → User se input leta hai.
- **raw_input()** → Sirf Python 2 me tha.
- **import** → Module ko program me use karne ke liye.

3.12 Operators

Definition

Operator ek **symbol** hota hai jo variables ya values par operation perform karta hai.

Jin values par operator kaam karta hai unhe **Operands** kehte hain.

Example

a = 10

b = 5

c = a + b

Yaha

- + → Operator
- a aur b → Operands

Types of Operators

Python me mainly 7 types ke operators hote hain.

1. Arithmetic Operators
2. Comparison (Relational) Operators
3. Assignment Operators
4. Bitwise Operators
5. Logical Operators
6. Membership Operators
7. Identity Operators

---

1. Arithmetic Operators

Definition

Arithmetic Operators ka use mathematical calculations ke liye hota hai.

Operator	Name	Example	Output
+	Addition	10+5	15
-	Subtraction	10-5	5
*	Multiplication	10*5	50
/	Division	10/5	2.0
%	Modulus	10%3	1
**	Exponent	5**2	25
//	Floor Division	10//3	3

Example

```
a=10
b=5


print(a+b)
```

```
print(a-b)
print(a*b)
print(a/b)
print(a%b)
print(a**2)
print(a//3)
```

Output

```
15
5
50
2.0
0
100
3
```

Difference Between / and //

```
print(10/3)
```

Output

```
3.333333
```

print(10//3)

Output

3

Real Life Example

- / → Exact result (calculator jaisa)
- // → Sirf integer part (decimal hata deta hai)

---

2. Comparison (Relational) Operators

Definition

Do values ko compare karte hain.

Result hamesha

- True
- False

me aata hai.

Operator	Meaning
==	Equal
!=	Not Equal
>	Greater
<	Less
>=	Greater than Equal
<=	Less than Equal

Example

```
a=10
b=5


print(a==b)
print(a!=b)
print(a>b)
print(a<b)
print(a>=b)
```

print(a<=b)

Output

False

True

True

False

True

False

Real Life Example

Teacher check karta hai

"Marks 33 se jyada hai?"

Ye comparison hai.

---

3. Assignment Operators

Definition

Variable me value store karne ke liye use hote hain.

Operator	Meaning
=	Assign
+=	Add and Assign
-=	Subtract and Assign
*=	Multiply and Assign
/=	Divide and Assign
%=	Modulus and Assign
**=	Power and Assign
//=	Floor Divide and Assign

Example

```
a=10


a+=5


print(a)
```

Output

15

Another Example

a=20

a//=3

print(a)

Output

6

Shortcut

a=a+5

same as

a+=5

---

4. Bitwise Operators

Definition

Bitwise Operators numbers ke **Binary Bits** par work karte hain.

Operator	Name
&	AND
|	OR
^	XOR
~	NOT
<<	Left Shift
>>	Right Shift

Example

```
a=5

b=3


print(a&b)
```

Output

```
1
```

Binary Explanation

5 = 0101

3 = 0011

AND

0001

=1

OR Example

print(5|3)

Output

7

Left Shift

print(5<<1)

Output

10

Right Shift

print(8>>2)

Output

2

Interview Point

Bitwise Operators mostly

- Networking
- Embedded System
- Operating System
- Encryption

me use hote hain.

---

5. Logical Operators

Definition

Conditions ko combine karte hain.

Operator	Meaning
and	Dono True
or	Koi ek True
not	Result ulta kar deta hai

Example

```
a=10
b=5

print(a>b and b>2)

print(a>b or b<2)

print(not(a>b))
```

Output

```
True
True
False
```

Real Life Example

Exam pass hone ke liye

Age>18 **AND**

Documents Complete

Dono true hone chahiye.

---

6. Membership Operators

Definition

Check karta hai ki koi value list ya string me present hai ya nahi.

Operator	Meaning
in	Present
not in	Present nahi hai

Example

name="Vijay"

```
print("V" in name)


print("z" in name)
```

Output

True

False

Another Example

```
numbers=[10,20,30]


print(20 in numbers)


print(40 not in numbers)
```

Output

True

True

Real Life Example

Attendance list me naam hai ya nahi.

---

7. Identity Operators

Definition

Ye compare karte hain ki do variables **same object** ko refer kar rahe hain ya nahi.

Operator	Meaning
is	Same Object
is not	Different Object

Example

a=10

b=10

print(a is b)

Output

True

Example

a=[1,2]

b=[1,2]

print(a==b)

print(a is b)

Output

True

False

Difference

==

→ Value compare karta hai.

is

→ Memory Location compare karta hai.

3.12.8 Operator Precedence

Definition

Ek expression me multiple operators ho to pehle kaunsa execute hoga, ye Operator Precedence decide karti hai.

Example

print(5+3*2)

Output

11

Pehle

3×2=6

Fir

5+6=11

Operator Precedence Order (High → Low)

Priority	Operators
1	**
2	+x -x ~x
3	* / // %
4	+ -
5	<< >>
6	&
7	^
8	`
9	== != < > <= >=
10	is, is not
11	in, not in
12	not
13	and
14	or

Real Life Example

School Assembly

Principal

↓

Teachers

↓

Students

Jaise sabki priority hoti hai,

waise hi Operators ki bhi priority hoti hai.

Chapter Summary

Is chapter me humne seekha:

- ✅ Python kya hai
- ✅ Python ki Features
- ✅ Python kaise Run karte hain
- ✅ Identifiers
- ✅ Reserved Keywords
- ✅ Variables
- ✅ Comments

- [x] Indentation
- [x] Multi-Line Statements
- [x] Suite
- [x] Quotes
- [x] print()
- [x] input()
- [x] import
- [x] Arithmetic Operators
- [x] Comparison Operators
- [x] Assignment Operators
- [x] Bitwise Operators
- [x] Logical Operators
- [x] Membership Operators
- [x] Identity Operators
- [x] Operator Precedence

Exam Important Definitions

Python

Python ek High-Level, Interpreted, Object-Oriented aur General Purpose Programming Language hai.

Identifier

Variable, Function, Class ya Object ka naam Identifier kehlata hai.

Keyword

Python ke Reserved Words ko Keywords kehte hain.

Variable

Memory me value store karne ke liye Variable use hota hai.

Comment

Program ko explain karne ke liye likha gaya text Comment kehlata hai.

Indentation

Python me spaces ya tabs ka use karke code block banana Indentation kehlata hai.

Multi-Line Statement

Jab ek statement ek se adhik lines me likha jaye to use Multi-Line Statement kehte hain.

Suite

Same indentation wale statements ke group ko Suite kehte hain.

---

print()

Output screen par dikhane ke liye use hota hai.

---

input()

User se input lene ke liye use hota hai.

---

import

Module ko program me use karne ke liye Import keyword ka use hota hai.

Operator

Operator ek symbol hota hai jo operands par operation perform karta hai.

Interview Questions

Q1. Python interpreted language kyu kehlati hai?

Kyuki code line by line execute hota hai aur pehle compile nahi hota.

Q2. Difference between == and is?

- == → Values compare karta hai.
- is → Memory location (object identity) compare karta hai.

Q3. Difference between / and //?

- / → Floating-point division.
- // → Floor division (decimal part hata deta hai).

Q4. Difference between in and is?

- in → Collection (list, string, tuple, etc.) me value present hai ya nahi check karta hai.
- is → Do variables same object ko refer kar rahe hain ya nahi check karta hai.

Q5. Python me comments kaise likhte hain?

- Single-line comment → #
- Multi-line text/docstring → Triple quotes ''' ''' ya """ """

📌 **Final Revision (1 Minute)**

- Python → High-Level, Interpreted Language
- Identifier → Variable/Function/Class ka naam
- Keyword → Reserved word
- Variable → Data store karta hai

- Comment → Code explain karta hai
- Indentation → Code block define karta hai
- Suite → Same indentation ka group
- print() → Output
- input() → Input
- import → Module use karna
- Arithmetic → Calculation
- Comparison → True/False comparison
- Assignment → Value assign/update
- Bitwise → Binary bits par operation
- Logical → and, or, not
- Membership → in, not in
- Identity → is, is not
- Operator Precedence → Execution order